

# Answer Key & Teacher Notes

Session 1 – Find the Percent of a Number • Session 2 – Compare Percents

## HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

|                                  |                                                                                                                                                                                    |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Warm-Up</b>                   | First 5–8 minutes of Session 1. Key: Unit Warm-Up Key.                                                                                                                             |
| <b>Prerequisite Practice A/B</b> | Before the lesson, for students who need the skill the session uses. Key in that folder.                                                                                           |
| <b>Guided Notes</b>              | During the lesson. Students fill in the blanks as you teach; answers below.                                                                                                        |
| <b>Practice Form A</b>           | In class, after the lesson. Two pages. The self-check code tells students if they are on track.                                                                                    |
| <b>Practice Form B</b>           | Same skills, new numbers: reteach, retake, small group or absent students.                                                                                                         |
| <b>Exit Ticket</b>               | Last 5 minutes. Two per page; cut in half.                                                                                                                                         |
| <b>Homework Packet</b>           | Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support. |

## SESSION 1 • TEACHER NOTES

Find the Percent of a Number

### TALK PROMPTS

- Why do you multiply by the percent written in hundredths instead of by the percent number itself?
- Before you multiply, how do you know 35% of 160 has to be less than 160?

### PARTNER TASK

Music Playlist and Book Pages. One partner writes the percent as a decimal and the equation; the other multiplies and checks that the part is less than the whole. Switch roles with Pizza Orders: 35% of 160 pizzas (56).

### WATCH FOR

- Multiplying by the percent number ( $30 \times 70$ ) instead of the decimal ( $0.30 \times 70$ ).
- Writing small percents with the wrong place value (6% as 0.6, 9% as 0.9).
- Dividing the whole by the percent ( $80 \div 25$  for 25% of 80) instead of multiplying by it.

## SESSION 2 • TEACHER NOTES

Compare Percents

### TALK PROMPTS

- Why does turning each record into a percent make the comparison fair?
- Friday's club had more members (65 vs. 60). Why does that not decide which turnout was better?

### PARTNER TASK

Club Days. One partner finds the Monday and Friday percents (75%, 60%); the other explains which day was better and why counts alone do not decide. Switch roles with Recycling.

### WATCH FOR

- Ranking records by the part alone (22 out of 40 "beats" 12 out of 15).
- Comparing the part to 100 instead of to its own whole (33 out of 55 called 33%).
- Missing a tie when two different fractions simplify to the same percent ( $34/40$  and  $51/60$ ).

## ITEM ANALYSIS

depth of knowledge (DOK) for each practice item

| Sheet              | Standard  | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
|--------------------|-----------|---|---|---|---|---|---|---|---|---|----|----|
| <b>S1 • Form A</b> | 6.RP.A.3c | 1 | 2 | 1 | 1 | 2 | 2 | 2 | 3 | 3 | 1  | 1  |
| <b>S1 • Form B</b> | 6.RP.A.3c | 1 | 2 | 1 | 1 | 2 | 2 | 2 | 3 | 3 | 1  | 1  |
| <b>S2 • Form A</b> | 6.RP.A.3c | 1 | 2 | 1 | 1 | 2 | 2 | 2 | 3 | 3 | 1  | 1  |
| <b>S2 • Form B</b> | 6.RP.A.3c | 1 | 2 | 1 | 1 | 2 | 2 | 2 | 3 | 3 | 1  | 1  |

Items before the last three are Practice; then Find the Error (DOK 3) and two Test Practice items.

## GUIDED NOTES

**part:** The piece of the whole that the **percent** tells about.

**whole:** The full amount. It comes right after the word “of.”

**equation:** A math sentence with an **equal** sign.

- Write the percent as a **decimal**:  $45\% = 0.45$  and  $6\% = 0.06$ .
- Multiply:  $\text{part} = \text{percent} \times \text{whole}$ .
- If the percent is less than 100%, the part is **less** than the whole.

**Example:** The whole is **140** tickets.  $45\% = 0.45$   $0.45 \times 140 = 63$  Answer: **63** student tickets

## YOU TRY

1. a. 12 b. 20 c. 27
2.  $0.15 \times 640 = 96$  absent;  $640 - 96 = 544$  present

## EXIT TICKET

1. a. 36 b. 30
2.  $0.06 \times 45 = \$2.70$
3. **C** (0.12)

## PRACTICE FORM A

1. a. 20 b. 23 c. 21 d. 7 e. 44 f. 36
2. Milk  $0.30 \times 200 = 60$ ; water  $0.55 \times 200 = 110$ ; juice  $100\% - 85\% = 15\%$ ,  $0.15 \times 200 = 30$ . Check:  $60 + 110 + 30 = 200$ .
3.  $0.25 \times 80 = 20$  songs
4.  $0.35 \times 160 = 56$  pizzas
5. a.  $0.65 \times 320 = 208$  pages b.  $320 - 208 = 112$  pages
6. No. Discount =  $0.20 \times 48 = \$9.60$ . Sam used the percent number as dollars.
7. a.  $0.40 \times 180 = 72$  peppers b.  $100\% - 75\% = 25\%$ ;  $0.25 \times 180 = 45$  seedlings
8. a.  $0.65 \times 1,200 = 780$  tickets b.  $780 \times \$18 = \$14,040$  c.  $1,200 - 780 = 420$  empty seats, which is 35% of the hall
9. 30% is 0.30, not 30. A part of \$70 must be less than \$70.  $0.30 \times 70 = \$21$ .
10. **B** (30)
11. **C** (0.45)

**Self-check word** DECIMAL

## PRACTICE FORM B

1. a. 16 b. 47 c. 18 d. 13 e. 28 f. 30
2. Milk  $0.20 \times 300 = 60$ ; water  $0.45 \times 300 = 135$ ; juice  $100\% - 65\% = 35\%$ ,  $0.35 \times 300 = 105$ . Check:  $60 + 135 + 105 = 300$ .
3.  $0.25 \times 60 = 15$  songs
4.  $0.35 \times 140 = 49$  pizzas
5. a.  $0.85 \times 240 = 204$  pages b.  $240 - 204 = 36$  pages
6. No. Discount =  $0.20 \times 65 = \$13$ . Ari used the percent number as dollars.
7. a.  $0.45 \times 160 = 72$  peppers b.  $100\% - 75\% = 25\%$ ;  $0.25 \times 160 = 40$  seedlings
8. a.  $0.85 \times 800 = 680$  tickets b.  $680 \times \$15 = \$10,200$  c.  $800 - 680 = 120$  empty seats, which is 15% of the hall
9. 9% is 0.09, not 0.9 (0.9 is 90%).  $0.09 \times 200 = 18$ .
10. **B** (32)
11. **C** (0.15)

**Self-check word** PERCENT

## GUIDED NOTES

**part-to-whole ratio:** Compares part of a group to the **whole** group.

**common whole:** The same total, like **100**, used for every group.

**compare:** To decide which amount is **greater** or less.

- Counts alone can be unfair when the **wholes** are different.
- Write each record as a **percent**, then compare the percents.

**Example:** Mia:  $21 \div 28 = 0.75 = 75\%$  Leo:  $16 \div 20 = 0.80 = 80\%$  Compare:  $80\% > 75\%$  Answer: **Leo** has the better rate, even though Mia made more shots.

## YOU TRY

1. a. 75% b. 72%
2. Ana 70%, Ben 68%. Ana has the better rate.

## EXIT TICKET

1. a. 90% b. 65%
2. Team A 75%, Team B 70%. Team A has the better record.
3. **A** (7 out of 10)

## PRACTICE FORM A

1. a. 90% b. 80% c. 75% d. 84% e. 80% f. 60%
2. Dara 75%, Ivan 65%, Nora 80%. Nora is the best shooter, even though she did not take the most shots.
3. a. A 85%, B 85% b. tie
4. a. Monday 75%, Friday 60% b. Monday
5. Rico  $9/12 = 3/4 = 75\%$ ; Tia  $15/24 = 5/8 = 62.5\%$ . Rico had the better rate.
6. No.  $33/55 = 3/5 = 60/100 = 60\%$ . Ben compared 33 to 100 instead of to 55.
7. Ella  $63/70 = 90\%$ ; Omar  $46/92 = 50\%$ ;  $90 - 50 = 40$  percentage points.
8. a. 80%, 80%, 75% b. Classes 1 and 2 tie for first; Class 3 is third c. Class 2 also had the most items to recycle, so its share (80%) matters, not its count of 40
9. Theo compared only the parts, not the parts to their wholes.  $12/15 = 80\%$  and  $22/40 = 55\%$ , so 12 out of 15 is the better score.
10. **C** (90%)
11. **A** (12 out of 15)

**Self-check word** COMPARE

## PRACTICE FORM B

1. a. 85% b. 60% c. 75% d. 76% e. 70% f. 70%
2. Amir 75%, Bea 60%, Cole 85%. Cole is the best shooter.
3. a. C 90%, D 90% b. tie
4. a. Tuesday 75%, Thursday 80% b. Thursday
5. Jada  $14/16 = 7/8 = 87.5\%$ ; Ken  $18/20 = 90\%$ . Ken had the better rate.
6. No.  $28/40 = 7/10 = 70/100 = 70\%$ . Mae compared 28 to 100 instead of to 40.
7. Lia  $72/80 = 90\%$ ; Sam  $33/60 = 55\%$ ;  $90 - 55 = 35$  percentage points.
8. a. 80%, 70%, 70% b. Class 1 first; Classes 2 and 3 tie c. Class 2 had 60 items to recycle, so 42 is only 70% of its total; Class 1 recycled a greater share
9. Rae compared counts with different totals.  $30/60 = 50\%$  and  $20/25 = 80\%$ , so 20 out of 25 is the better record.
10. **C** (80%)
11. **A** (9 out of 10)

**Self-check word** RECORD

## HOMEWORK PACKET • ANSWERS

*same answers for the core packet and Level 1*

**SHEET 1 · FIND THE PERCENT OF A NUMBER**

1. a. 9 b. 48 c. 21 d. 36 e. 45 f. 30
2. Missing: 72, 108, 144.  $40\%$  of  $180 = 72$  students.
3. a.  $0.25 \times 64 = \$16$  b.  $\$64 - \$16 = \$48$
4.  $10\%$  of  $40 = 4$  and  $5\% = 2$ , so  $15\% = 6$  players ( $0.15 \times 40 = 6$ ).
5. They are equal:  $0.3 \times 90 = 27$  and  $0.9 \times 30 = 27$ . Both equal  $30 \times 90 \div 100$ , so switching the numbers gives the same amount.

**SHEET 2 · COMPARE PERCENTS**

1. a.  $70\%$  b.  $65\%$  c.  $75\%$
2. Ava  $45\%$ , Ben  $48\%$ , Cruz  $44\%$ . Ben has the best record.
3. Rosa:  $34/40 = 85/100 = 85\%$ . Kai:  $42/50 = 84/100 = 84\%$ . Rosa did better, even though Kai spelled more words.
4. A, C ( $75\%$  and  $72\%$ ;  $17$  out of  $25 = 68\%$ ,  $7$  out of  $10 = 70\%$  exactly)
5. Equal rates come from multiplying, not adding the same amount.  $18/24 = 3/4 = 75\%$  and  $24/30 = 4/5 = 80\%$ , so the rates are not equal;  $24$  out of  $30$  is better.
6. **C** ( $33$  out of  $50$ )
7. **B** (A won  $75\%$ , B won  $60\%$ . A did better.)